

Lab 4 - Bayesian Networks

Group 3 - Subgroup 13

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Tasks for part II

- 5.
 - a) The risk of meltdown in the power plant if no observations have been made is 0.02578, and it is 0.03472 when icyWeather is set to true.
 - b)
 - * $P(\text{Meltdown} \mid \text{PumpFailureWarning} \cap \text{WaterLeakWarning}) = 0.14535$
 - * $P(\text{Meltdown} \mid \text{PumpFailure} \cap \text{WaterLeak}) = 0.2$
 - c) Getting accurate numbers for conditional probabilities can be difficult when there are not enough events to observe. It is also tough to understand how different factors influence each other. In the plant model, estimating probabilities like $P(\text{Meltdown} \mid \text{PumpFailure}, \text{WaterLeak})$ can be challenging because meltdowns don't happen often and there isn't much data.
 - d)
 - * If icyWeather changes to Temperature, the domain could be categories like cold, mild, hot or numerical as degrees.
 - For categories, $P(\text{WaterLeak} \mid \text{Temperature})$ changes with each label: cold might increase leak chances, hot decrease.
 - For numerical, the relationship is more complex, showing how leak chances vary with temperature changes.
- 6
 - a) A probability table in a Bayesian network shows the chances of each outcome for a node, based on the states of its parent nodes. It helps to predict what happens next based on what we already know.

- b)

- A joint probability distribution shows the probability of different events happening together.
- The chain rule in probability allows us to break down the joint probability of a sequence of events into a product of conditional probabilities.
- Using the chain rule, the entry where all variables are false can be found as the following:

- * Variables:

- M for Meltdown
 - PFW for PumpFailureWarning
 - PF for PumpFailure
 - WLW for WaterLeakWarning
 - WL for WaterLeak
 - IW for IcyWeather

- * Values from Applet:

- $P(IW = F) = 0.95$
 - $P(WL = F|IW = F) = 0.90$
 - $P(PF = F) = 0.90$
 - $P(M = F|PF = F, WL = F) = 0.99$
 - $P(PFW = F|PF = F) = 0.95$
 - $P(WLW = F|WL = F) = 0.95$

- * Chain rule:

- $P(M = F, PFW = F, PF = F, WLW = F, WL = F, IW = F) =$
 - $P(IW = F) \times$
 - $P(WL = F|IW = F) \times$
 - $P(PF = F) \times$
 - $P(M = F|PF = F, WL = F) \times$
 - $P(PFW = F|PF = F) \times$
 - $P(WLW = F|WL = F)$

- * Plug in the values:

$$0.95 \times 0.90 \times 0.90 \times 0.99 \times 0.95 \times 0.95 \approx 0,687$$

- **Is this a common state for the nuclear plant to be in?**

Yes. The chance to be in this state is about 70% according to the calculation above so it's common.

- c)
 - The probability of a meltdown if you know that there is both a water leak and a pump failure is: $P(M = T \mid PF, WL) = 0.2$.
 - Knowing the state of other variables like IcyWeather, or WaterLeakWarning doesn't change the probability of a Meltdown because Meltdown depends only directly on Pump Failure and Water Leak. Once it's known that there is both a pump failure and a water leak, the other variables don't directly affect the Meltdown.
- d)
 - Values from Applet:
 - * $P(IW = F) = 0.95$
 - * $P(WL = F \mid IW = F) = 0.90$
 - * $P(WLW = F \mid WL = F) = 0.95$
 - * $P(PF = F) = 0.90$
 - * $P(PF = T) = 0.10$
 - * $P(PFW = F \mid PF = F) = 0.95$
 - * $P(PFW = F \mid PF = T) = 0.10$
 - * $P(M = T \mid PF = T, WL = F) = 0.15$
 - * $P(M = F \mid PF = T, WL = F) = 0.85$
 - * $P(M = T \mid PF = F, WL = F) = 0.001$
 - * $P(M = F \mid PF = F, WL = F) = 0.999$

– **For Meltdown = True:**

* For Meltdown being true, sum the contributions from each state of Pump Failure:

$$\begin{aligned}
P(M = T \mid PFW = F, WL = F, WLW = F, IW = F) = & \\
& (P(M = T \mid PF = T, WL = F) \times \\
& P(PFW = F \mid PF = T) \times \\
& P(PF = T) \times \\
& P(WLW = F \mid WL = F) \times \\
& P(WL = F \mid IW = F) \times \\
& P(IW = F)) \\
& + \\
& (P(M = T \mid PF = F, WL = F) \times \\
& P(PFW = F \mid PF = F) \times \\
& P(PF = F) \times \\
& P(WLW = F \mid WL = F) \times \\
& P(WL = F \mid IW = F) \times \\
& P(IW = F))
\end{aligned}$$

* Plug in the values:

$$\begin{aligned}
& (0.15 \times 0.10 \times 0.10 \times 0.95 \times 0.90 \times 0.95) \\
& + \\
& (0.001 \times 0.95 \times 0.90 \times 0.95 \times 0.90 \times 0.95) \\
& = 0.001218375 + 0.00069447375 \approx 0.00191
\end{aligned}$$

– **For Meltdown = False:**

$$\begin{aligned}
P(M = F | PFW = F, WL = F, WLW = F, IW = F) = \\
& (P(M = F | PF = T, WL = F) \times \\
& \quad P(PFW = F | PF = T) \times \\
& \quad P(PF = T) \times \\
& \quad P(WLW = F | WL = F) \times \\
& \quad P(WL = F | IW = F) \times \\
& \quad P(IW = F)) \\
& + \\
& (P(M = F | PF = F, WL = F) \times \\
& \quad P(PFW = F | PF = F) \times \\
& \quad P(PF = F) \times \\
& \quad P(WLW = F | WL = F) \times \\
& \quad P(WL = F | IW = F) \times \\
& \quad P(IW = F))
\end{aligned}$$

* Plug in the values:

$$\begin{aligned}
& (0.85 \times 0.10 \times 0.10 \times 0.95 \times 0.90 \times 0.95) \\
& + \\
& (0.999 \times 0.95 \times 0.90 \times 0.95 \times 0.90 \times 0.95) \\
& = 0.006904125 + 0.6937792762 \approx 0.700
\end{aligned}$$

– **Normalization:**

* Normalization factor:

$$\alpha = \frac{1}{(0.00191 + 0.700)} \approx 1,424$$

* Normalized probability of a meltdown:

$$P(M = T) = 0.00191 \times 1.424 \approx 0.0027$$

So the answer is 0.0027 or 0.27%.

Task for part III

- 2.

- a) Surviving probability dropped from around 0.99 to around 0.98 given that the radio failed to work.

$$(P(\textit{Survive} \mid \textit{Radio} = F) \approx 0.98)$$

- b)

- * If there is no observation about the bicycle, the probability is:

$$P(\textit{Survive}) = 0.99544$$

- * If the bicycle works, the probability becomes:

$$P(\textit{Survive} \mid \textit{bicycle}) = 0.99594$$

- * If the bicycle does not work, the probability becomes:

$$P(\textit{Survive} \mid \textit{bicycle} = F) = 0.99090$$

- c)

- * It is possible to model any function in propositional logic with Bayesian Networks. Exact inference in probabilistic models is considered NP-hard, meaning it typically requires computational resources that grow exponentially with the size of the problem which makes it impractical for large datasets or complex models.
- * Alternatives to exact inference:
 - **Approximation algorithms:** These simplify calculations to save time, such as Monte Carlo methods.
 - **Topological structural constraints:** Modifying the model's structure to reduce complexity.

Task for part IV

- 2.

- a) A new pump could compensate since Mr H.S. behaviour depends on the PumpFailureWarning which depends on the PumpFailure, and a good pump would less likely have a failure.
- b) We added a new node that represents any warning, combining Pump Failure Warning and Water Leak Warning using a logical OR.

- c) In a Bayesian Network model of a person, we usually think they always act the same way. But people can act differently when they are hungry or tired. For example, we model reactions to warnings like they are always the same, but this is not very true because humans are more complicated.
- d) A Bayesian Network could model IcyWeather over consecutive days by linking nodes from one day to the next. This would mean that icy conditions on one day increase the likelihood for the following day which allows the tracking of weather patterns over a sequence of days.